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High Performance Processors and Systems

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Name
Last Name

Problem 1 (50%)	
Problem 2 (50%)	
Total (10 points)	

Problem 1

1. To implement cache coherency one of the protocols is the MESI concurrency protocol implemented for a write-invalidate write-back cache. Describe briefly the algorithm (also in term of graph) and complete the following table.

State	Cache up to date?	Memory up to date?	Others have a copy?	Cache can respond other's reads?
Modified				
Exclusive				
Shared				
Invalid				

2. Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the MESI protocol is used, with write-back caches, write-allocate, and invalidation of other caches on write (instead of updating the value in the other caches).

Time	After Operation	P1 cache block state	P2 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P1: read block 1	Exclusive (1)	Invalid (0)	Yes	Yes
1	P2: read block 0				
2	P1: write block 0				
3	P2: read block 2				
4	P1: read block 1				
5	P2: write block 1				
6	P1: read block 1				
7	P2: write block 1				
8	P1: read block 0				
9	P2: write block 1				
10	P2: write block 1				
11	P1: read block 1				
11	P2: write block 1				
12	P1: read block 1				
13	P2: read block 1				

Problem 2

Assume that a given 5 stage pipeline MIPS architecture is working with a clock cycle equal to 2 ns. Using the MIPS code shown below,

I1: add \$s3, \$s2, 2
I2: sub \$s4, \$s3, \$s1
I3: add \$s5, \$s4, \$s1
I4: lw \$s6, 4(\$s4)
I5: sub \$s7, \$s4, \$s6

- Draw the pipeline schema showing all the RAW data conflicts. Solve the conflicts without using rescheduling and path forwarding.
- Reschedule the instructions to reduce the stalls; Draw the pipeline schema showing all the RAW data conflicts.
- The forwarding paths have been included in the pipeline. Start from the code in (a) and draw the pipeline schema showing all the forwarding paths that have to be used to solve the conflicts.

Answer

- a) Show all the RAW data conflicts

Istr	CK1	CK2	CK3	CK4	CK5	CK6	CK7	CK8	CK9	CK10	CK11	CK12	CK13
I1	IF	ID	EX	MEM	WB								
I2		IF	ID	EX	MEM	WB							
I3			IF	ID	EX	MEM	WB						
I4				IF	ID	EX	MEM	WB					
I5					IF	ID	EX	MEM	WB				

CPU without forwarding

[illegible]

b) Rescheduled code:

CPU without forwarding

[illegible]

c) CPU with the forwarding paths

[illegible]

